

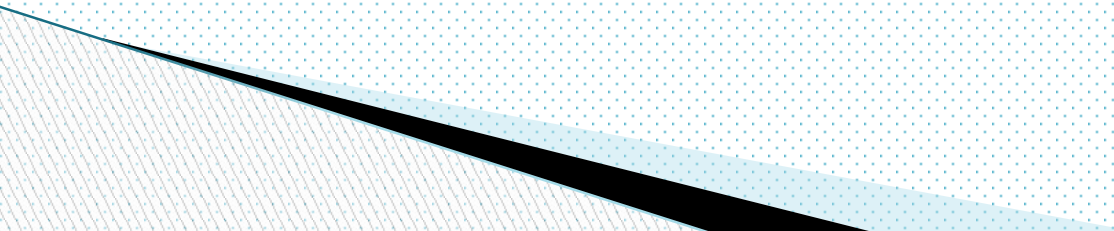
Regression and Correlation Analysis

7
(3)

*Introduction to
Rank Correlation*

7.3 Rank correlations

In studying the correlation coefficient we have discussed the correlation with its properties and also the calculation of *Karl Pearson's* correlation coefficient. In correlation coefficient, it is assumed that both characteristics are measurable.



7.3 Rank correlations

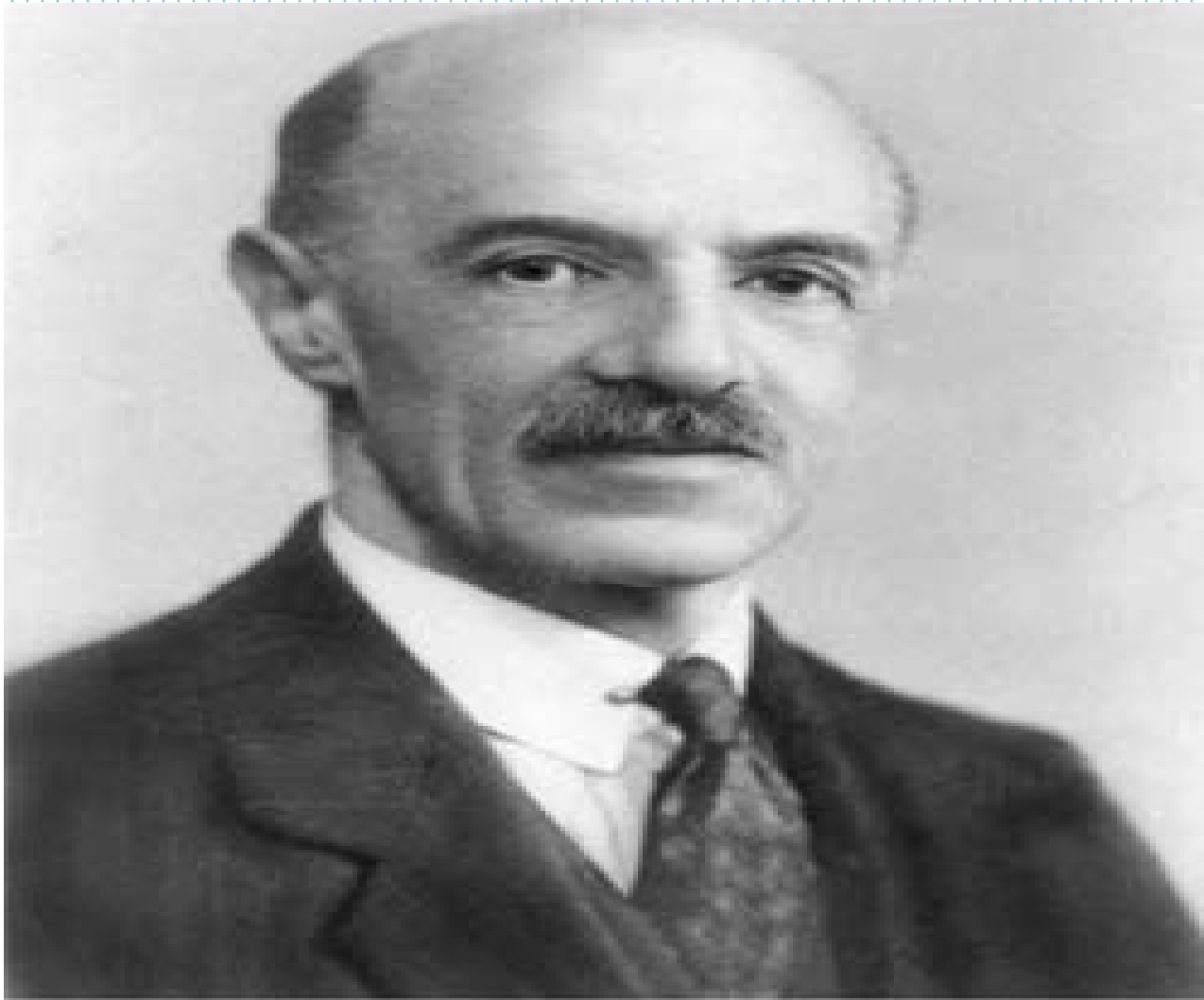
Sometimes characteristics are not measurable but ranks may be given to individuals according to their qualities. In such situations rank correlation is used to know the association between two characteristics. In this unit, we will discuss the rank correlation and calculation of rank correlation coefficient.



7.3 Rank correlations

- ▶ The Spearman correlation coefficient is to be used to measure rank correlation (statistical dependence of ranking between two variables).
- ▶ Named after Charles Spearman, it is often denoted by the Greek letter ' ρ ' (rho) and is primarily used for data analysis.

7.3 Rank correlations



Charles Edward Spearman (1863 - 1945)

7.3 Rank correlations

It measures the strength and direction of the association between two ranked variables.

The formula for Spearman correlation coefficient:

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} = R_r = r_s,$$

where n - number of data points of the two variables

d_i - difference in ranks of the “ i ^{*t*} *h*” element

7.3 Rank correlations

Example 1:- The following table shows the number of items produced and the cost incurred in producing them (in birr).

Number of items produced	Cost incurred (in birr)
4	15
5	18
6	18
8	20
9	22

7.3 Rank correlations

Use the Spearman coefficient of correlation and compare your result with the one you had using the *Karl Pearson's correlation coefficient*.

Number of items produced	Rank	Cost incurred (in birr)	Rank	d_i	d_i^2
4	5	15	5	0	0
5	4	18	3.5	0.5	0.25
6	3	18	3.5	-0.5	0.25
8	2	20	2	0	0
9	1	22	1	0	0
					$\Sigma d_i^2 = 0.5$

7.3 Rank correlations

$$\begin{aligned}\rho &= 1 - \frac{6\sum_i d_i^2}{n(n^2 - 1)} = 1 - \frac{6(0.5)}{5(5^2 - 1)} = 1 - \frac{3}{5(24)} \\ &= 1 - \frac{3}{120} = 1 - \frac{1}{40} = 0.975\end{aligned}$$

7.3 Rank correlations

Example 2: The following table refers to the marks (out of 50) of 5 randomly taken students in DRBSE2202 in two different courses: Communicative English and Math for Natural Sciences. Check the strength of the relationship of the two courses.

Math for Natural Sc.	35	23	47	17	43
Communicative English	30	33	45	23	49

7.3 Rank correlations

Solution:

Math	Rank	Eng.	Rank	d_i	d_i^2
35	3	30	4	-1	1
23	4	33	3	1	1
47	1	45	2	-1	1
17	5	23	5	0	0
43	2	49	1	1	1
					$\sum d_i^2 = 4$

$$\begin{aligned}\rho &= 1 - \frac{6\sum_i d_i^2}{n(n^2 - 1)} = 1 - \frac{6(4)}{5(5^2 - 1)} = 1 - \frac{24}{5(24)} \\ &= 1 - \frac{24}{120} = 1 - 0.2 = 0.8\end{aligned}$$

7.3 Rank correlations

Example 3: In a study of the relation between students' grades in Artificial Intelligence and Computer Graphics, the following results were found for six students. Find the Spearsman correlation coefficient and interpret your result.

Artificial Inteligence	D	B	A	B	D	D
Computer Graphics	C	C	B	A	C	F

7.3 Rank correlations

Solution:

AI	Rank	Comp. Graphics	Rank	d_i	d_i^2
D	5	C	4	1	1
B	2.5	C	4	-1.5	2.25
A	1	B	2	-1	1
B	2.5	A	1	1.5	2.25
D	5	C	4	1	1
D	5	F	6	-1	1
					$\sum d_i^2 = 8.5$

7.3 Rank correlations

$$\begin{aligned}\rho &= 1 - \frac{6\sum_i d_i^2}{n(n^2 - 1)} = 1 - \frac{6(8.5)}{6(6^2 - 1)} = 1 - \frac{51}{6(35)} \\ &= 1 - \frac{51}{210} = 1 - \frac{51}{210} = 0.757\end{aligned}$$

Therefore, we can conclude that there is a strong degree of association (or agreement) between the grades of the two subjects.

7.3 Rank correlations

Example 4:- At the Deepdale 'Best of British Pie' competition two judges award marks for nine different pies as follows:

Jude 1	Judge 2
18	7
24	18
23	9
13	4
27	17
19	8
30	29
10	8
20	10

7.3 Rank correlations

Using Spearman's rank correlation coefficient, find the measure of the degree of agreement between the two judges.

7.3 Rank correlations

Judge 1	Rank 1	Judge 2	Rank 2	d_i	d_i^2
18	7	7	8	-1	1
24	3	18	2	1	1
23	4	9	5	-1	1
13	8	4	9	-1	1
27	2	17	3	-1	1
19	6	8	6.5	-0.5	0.25
30	1	29	1	0	0
10	9	8	6.5	2.5	6.25
20	5	10	4	1	1
					$\sum d_i^2 = 12.5$

7.3 Rank correlations

$$\begin{aligned}\rho &= 1 - \frac{6\sum_i d_i^2}{n(n^2 - 1)} = 1 - \frac{6(12.5)}{9(9^2 - 1)} = 1 - \frac{75}{9(80)} \\ &= 1 - \frac{75}{720} = 1 - \frac{5}{48} = 0.895\end{aligned}$$

Therefore, we can conclude that there is a very strong degree of association (or agreement) between the two judges.

7.3 Rank correlations

Example 5:- An investigation was conducted by a company on the value of various assessment methods for recruiting employees. The data are shown in this table.

7.3 Rank correlations

Employee	Educational Test Score	Assessment Score by Personnel Officer
A	9	12
B	10	14
C	15	16
D	14	15
E	16	17
F	11	10
G	12	11
H	17	18

7.3 Rank correlations

This is based on 8 employees, giving their educational test scores, together with an assessment score by the Personnel Officer of their ability one year after joining the company.

7.3 Rank correlations

- i) Rank each employee in terms of Educational Test score and Assessment score by the Personnel Officer.
- ii) Hence for these scores calculate, to 2 decimal places, the Spearman rank correlation coefficient.

Any Questions



Any Questions



Thank
you